

# Attractor networks

## Continuous attractors and Hopfield nets

Computational Neuroscience

University of Bristol

Adapted from C O'Donnel and L Aitchison

mrule

QIFI.  
Email about  
fixes.

### Learning outcomes:

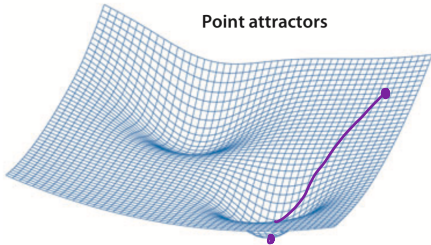
- ▶ Continuous attractors
  - Head-direction cell attractors
  - Grid-cell attractors
- ▶ Hopfield nets
  - Be able to explain what the Hopfield energy function is,
  - how it relates to stored patterns, and
  - how it evolves under asynchronous updates
  - Hebbian plasticity and the Hopfield network

## *Attractor networks*

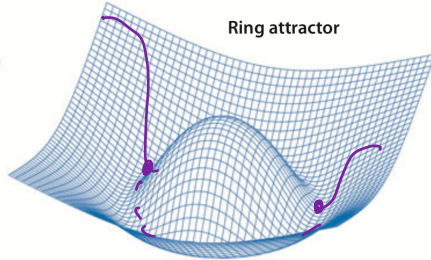
# Attractor networks

## a Energy landscape

Point attractors



Ring attractor



Knierim, J.J., Zhang, K., 2012. *Annu. Rev. Neurosci.* 35, 267–285.

Attractor networks are recurrent neural networks where the internal dynamics evolves towards a stable pattern.

## *Attractor networks*

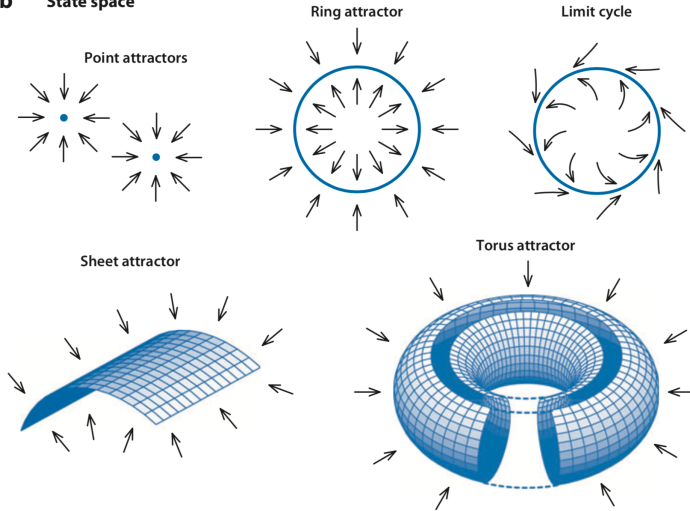
What's vague about the previous slide?

Attractor networks are ...

- ▶ recurrent neural networks where the internal dynamics evolves towards a stable pattern.

*Continuous attractors*

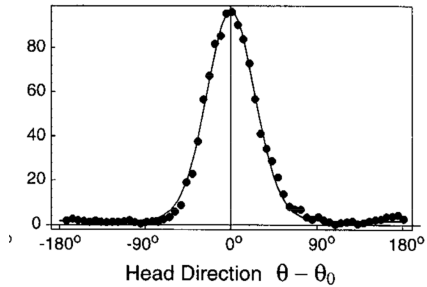
**b** State space



*Continuous attractors in navigation*

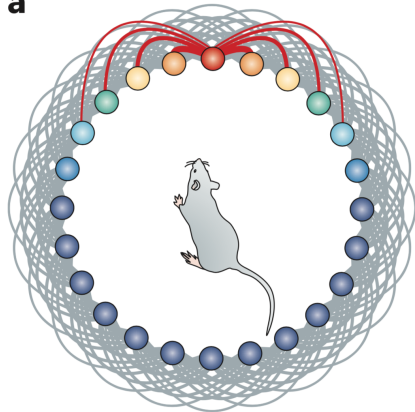


## Continuous attractors in subiculum



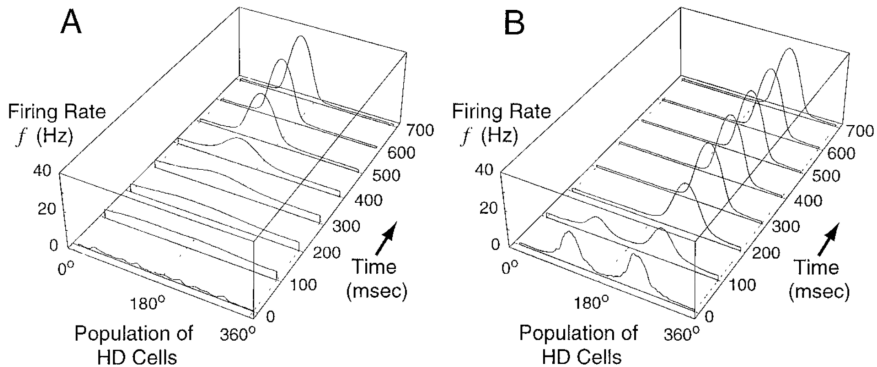
McNaughton, et al. (2006). *Nat Rev Neurosci* 7, 663–678

**a**



Knierim, J.J., Zhang, K., 2012. *Annu. Rev. Neurosci.* 35, 267–285.

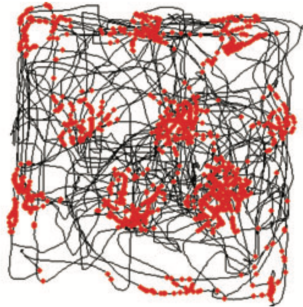
## Continuous attractors in subiculum



Knierim, J.J., Zhang, K., 2012. *Annu. Rev. Neurosci.* 35, 267–285.

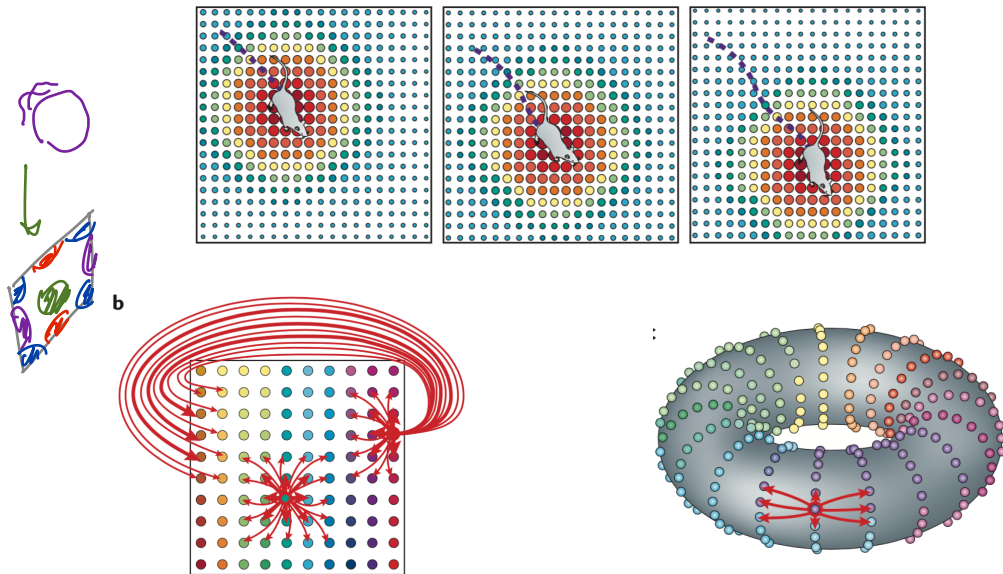
## *Continuous attractors in entorhinal cortex*

### **Grid cell**

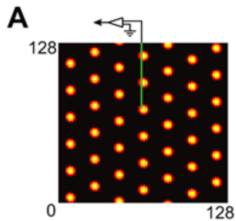


*McNaughton, et al. (2006). Nat Rev Neurosci 7, 663–678*

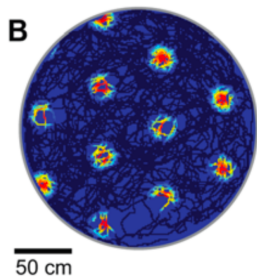
## Continuous attractors in entorhinal cortex



## Continuous attractors in entorhinal cortex



Network activity

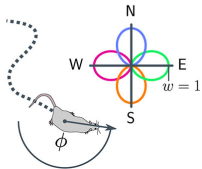


Single cell activity as a function of location

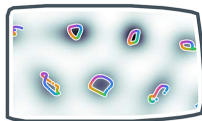
## Then there's this one cell...

(a) Head-direction weights

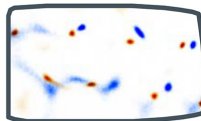
$$w = \max[0, \cos(\phi - \phi_0)]^2$$



(c) Tracked peaks



(d) Northwards (blue) vs southwards (orange) vs southwards (orange)

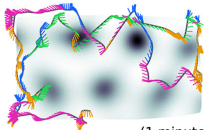


Eastwards (green) vs westwards (pink)



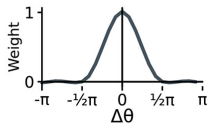
50 cm

(b) Head direction



(1 minute excerpt)

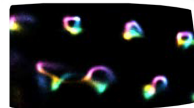
(e) Head-direction kernel



(f) Joint rate map



(g) Joint peak density



E S W N

## *Hopfield nets*

## Hopfield network

### Recurrent Neural Network (RNN)

- ▶ Discrete time ( $\Delta$  steps, not  $d/dt$ )
- ▶ Discrete states  $x_{i,t} \in \{-1, 1\}$
- ▶ McCulloch-Pitts (linear threshold) units

$$s = 2(?) - 1 \quad \{-1, 1\}$$

### Proposed by John Hopfield (1982)

- ▶ Basic model of associative memory recall
- ▶ **2024 Nobel Prize in Physics**

States evolve dynamically, typically toward some “attractor” state. A synaptic plasticity rule can imprint attractors in the network weights.



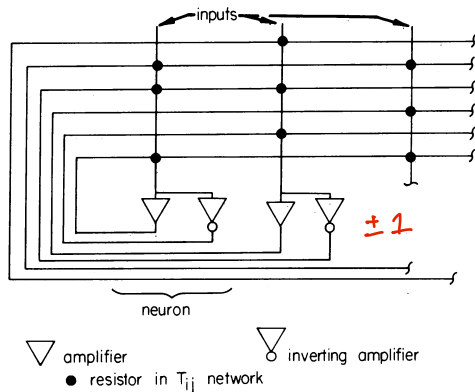
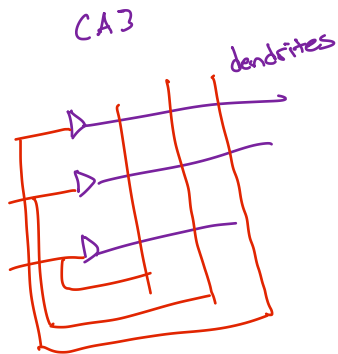


FIG. 2. An electrical circuit that corresponds to Eq. 5 when the amplifiers are fast. The input capacitance and resistances are not drawn. A particularly simple special case can have all positive  $T_{ij}$  of the same strength and no negative  $T_{ij}$  and replaces the array of negative wires with a single negative feedback amplifier sending a common output to each "neuron."

## Hopfield networks

Weights (in this course)

- ▶ Always symmetric,  $w_{ij} = w_{ji}$
- ▶ No self edges (no autapses),  $w_{ii} = 0$

State dynamics:

- ▶ Synchronous:

$$\mathbf{x}_{t+1} \leftarrow \text{sign}[\mathbf{W}\mathbf{x}_t - \boldsymbol{\vartheta}]$$

- ▶ Asynchronous: Pick a unit index  $i$ , then

$$x_{i;t+1} \leftarrow \text{sign}[\sum_{ij} w_{ij}x_{t;j} - \vartheta_i]$$

## Hopfield networks

States evolve to minimise an “energy”:

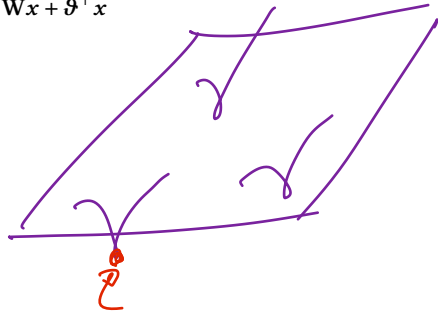
$$E = -\frac{1}{2} \sum_{ij} w_{ij} x_i x_j + \sum_i \vartheta_i x_i = -\frac{1}{2} \mathbf{x}^\top \mathbf{W} \mathbf{x} + \boldsymbol{\vartheta}^\top \mathbf{x}$$

Nuances:

- ▶ The synchronous update can get stuck in cycles
- ▶ There are tweaks to prevent this
- ▶ The asynchronous update does not get stuck like this

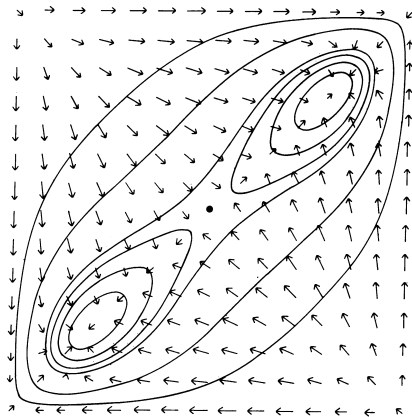
And you need

- ▶ Weights symmetric,  $w_{ij} = w_{ji}$
- ▶ No self edges (no autapses),  $w_{ii} = 0$



A stored memory is an energy  
Local minimum.





$[+1, +1]$

$[-1, -1]$

### Energy 'Landscape'

Local minima  $\rightarrow$  'attractors'

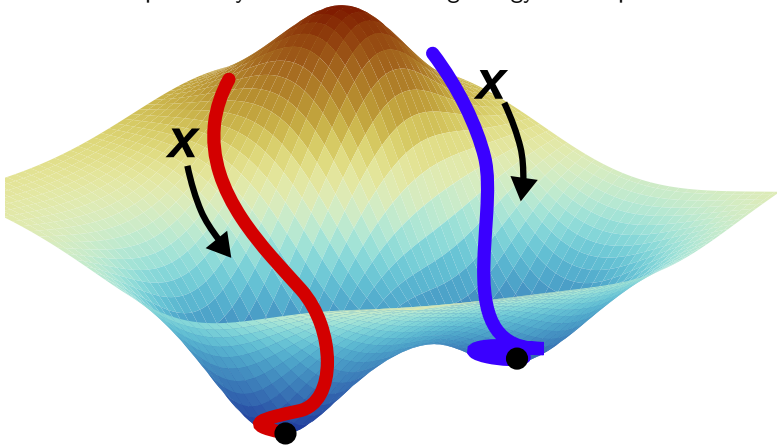
The network can do pattern completion: retrieving the full pattern from a partial cue.

The network we study here has capacity (maximum # stored attractors) of  $\sim 0.14N$ .

FIG. 3. An energy contour map for a two-neuron, two-stable-state system. The ordinate and abscissa are the outputs of the two neurons. Stable states are located near the lower left and upper right corners, and unstable extrema at the other two corners. The arrows show the motion of the state from Eq. 5. This motion is not in general perpendicular to the energy contours. The system parameters are  $T_{12} = T_{21} = 1$ ,  $\lambda = 1.4$ , and  $g(u) = (2/\pi)\tan^{-1}(\pi\lambda u/2)$ . Energy contours are 0.449, 0.156, 0.017,  $-0.003$ ,  $-0.023$ , and  $-0.041$ .

Hopfield, PNAS 1984

Pattern completion dynamics ~ descending energy landscape



## *Memorise me for the exam*

Hopfield network energy function:

$$E = -\frac{1}{2} \sum_{ij} w_{ij} x_i x_j + \sum_i \vartheta_i x_i = -\frac{1}{2} \mathbf{x}^\top \mathbf{W} \mathbf{x} + \boldsymbol{\vartheta}^\top \mathbf{x}$$





## Learning attractors in Hopfield networks

Consider a Hopfield network with  $N$  units (neurons)

We have a set  $\mathcal{X}$  of  $P$  patterns  $\mathbf{x}_a = \{x_{1;a}, x_{2;a}, \dots, x_{N;a}\}^\top$  (column vectors), where

- ▶ each  $x_{i;a} \in \{-1, 1\}$ , and
- ▶  $a \in 1 \dots P$  indexes over patterns
- ▶  $i \in 1 \dots N$  indexes over neurons

Define zero-mean patterns

$$\tilde{x}_{i;a} = x_{i;a} - \underbrace{\frac{1}{N} \sum_{i=1}^N x_{i;a}}_{\mu_a}$$

$$\tilde{\mathbf{x}}_a = \mathbf{x}_a - \mu_a$$

Store desired attractor states into the synaptic weights using a learning rule:

$$w_{ij} = \begin{cases} \frac{1}{P} \sum_{a=1}^P \tilde{x}_{i;a} \tilde{x}_{j;a} & \text{if } i \neq j \\ 0 & \text{if } i = j \end{cases} \quad (1)$$

$$\mathbf{W} = \langle \tilde{\mathbf{x}} \tilde{\mathbf{x}}^\top \rangle_{\mathcal{X}} - \mathbf{I}$$

Surprisingly ok model of CA3 *Symmetric STDP*!

### Hebbian plasticity

***When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased.***—Donald Hebb (1949)

***“neurons that fire together wire together”***—Dr. Carla Shatz (c. 1992)

To a first approximation:

$$\Delta w_{ij;t} \propto x_{i;t} x_{j;t} \quad (i \neq j)$$

Verify (for  $\vartheta = 0$ ), scaling of weights  $c\mathbf{W}$  by a positive constant does not change the dynamics, so we can present patterns sequentially  $t := a$  and use a variant of Equation (1) that is quite similar to Hebbian learning:

$$\mathbf{W}_{t+1} \leftarrow \mathbf{W}_t + (\tilde{\mathbf{x}}_t - \mu_t)(\tilde{\mathbf{x}}_t - \mu_t)^\top$$

Active research area, watch this space

- ▶ Modern Hopfield networks
- ▶ Dense associative memory
- ▶ Hopfield networks are all you need (2020)

We covered

- ▶ What attractor networks are
- ▶ Continuous attractors
  - Ring attractors for head direction, heading
  - Grid attractors for location
- ▶ Hopfield networks
  - Definition
  - Update rule
  - Energy function
  - Weight learning

Discuss coursework problem with students and consider working past exam problems live

$$x_1 = (-1, -1, 1)$$

$$x_2 = (1, 1, 1)$$

$$x_1 x_1^T - I = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix}$$

$$x_2 x_2^T - I = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

(sum up)

$$\begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\frac{1}{2}$$

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$W = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$x_n = (+1 \ +1 \ -1)^T$$

$$\theta = 1$$

$$Wx_n - \theta = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} - 0 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} - \theta = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

Sign  $x_{n+1} = (1, 1, -1)$   
 $\uparrow$  because  $\geq 1$  is 1 by defn.

















*end*





